

# VIPUL DINESH

## ROBOTICS ENGINEER

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### SUMMARY

A Mechatronics Engineer focused on robotics hardware and systems integration - mechanical design, actuation, embedded firmware (ESP32/STM32), and ROS2 control. Two-time International Rover Challenge winner who takes electromechanical systems from CAD and FEA through fabrication to reliable, deployable prototypes.

### EDUCATION

**Manipal Institute of Technology** — B. Tech, Mechatronics Engineering (Minor Specialisation in Robotics) *Manipal, India · Nov 2026*  
CGPA 9.68/10 · Branch Topper 2022–23 & 2023–24 · Top 5% Merit Scholarship (all four years)

### TECHNICAL SKILLS

**Mechanical Design & Analysis:** CAD (SolidWorks, Fusion 360, Creo), FEA (ANSYS), Mechanism Design (4-bar / 5-bar linkages, cycloidal drives), tolerance analysis

**Actuation & Embedded:** ESP32, STM32, motor drivers (DM542, TB6600), stepper & servo sizing via torque analysis, comms protocols (I2C, SPI, UART, CAN), sensors (IMU, encoders, LiDAR), LiPo battery & power systems

**Robotics & Software:** ROS2 (Gazebo, RViz, Nav2), Raspberry Pi, inverse kinematics, PID control, MATLAB / Simulink, C/C++, Python, Git, Linux, Hamlib / rotctld, GPredict

**Manufacturing:** FDM 3D printing & printer optimization, CAM, rapid prototyping, mechanical assembly & integration

### EXPERIENCE

**GiftedMinds** — Mechatronics R&D Intern *Bengaluru · Dec 2025 – Present*

- Designed and built a 12-DOF, ~8 kg quadruped robot that achieves stable walking and trotting gaits at 0.3 m/s, by deriving analytic inverse kinematics and implementing ROS2-based locomotion control on a Raspberry Pi 5.
- Engineered a dual-axis (azimuth–elevation) satellite tracker with sub-degree pointing accuracy and full-sky coverage (Az  $\pm 180^\circ$ , El 0– $90^\circ$ ) — enough to track any LEO pass or point to any GEO satellite — by sizing steppers via torque analysis and building a full-float bearing assembly for load distribution.
- Wrote ESP32 firmware that drives both axes simultaneously (non-blocking) while maintaining a live Wi-Fi/GPredict client link and supporting OTA updates, parsing Hamlib-format commands into coordinated step motion.
- Taught and mentored ~50 students over 6 months in weekend mechatronics sessions (CAD, robotics, control systems) — a recurring part of the role, with many going on to build and pitch complete products.

**Space Applications Centre, ISRO** — Research Project Intern *Ahmedabad · May – Jul 2025*

- Characterized a 6-DOF Stewart platform (hexapod) capable of sub-100  $\mu\text{m}$  linear and sub-10 arcsec angular accuracy for precision alignment of optical components in a satellite telescope.
- Supported thermal-vacuum (TVAC) and vibration qualification cycles, validating platform performance under launch and space-environment conditions.

**Mars Rover Manipal** — Mechanical Design & Manufacturing Head *Nov 2022 – Sep 2025*

- Led mechanical design and manufacturing to back-to-back International Rover Challenge (Host - SPROS) wins in 2024 and 2025, Top 10 at the European Rover Challenge 2025 (Host- European Space Agency), and 18th overall / 3rd in Science Mission at the University Rover Challenge 2024 (Host - The Mars Society).
- Developed a 3D-printed in-situ science module — a dual-auger leadscrew system that drills 10 cm sub-surface with 6 integrated sensors, plus a cache mechanism and fluid delivery for soil analysis — doubling the augers (1 to 2) and optimising the system for up to 100% faster sampling and sensor cycles.
- Directed design and fabrication of core subsystems — 4-bar robotic arm, cycloidal drives, modular chassis, 3D-printed wheels, and an optimized 5-bar suspension; led technical recruitment across 1000+ applicants.

**AIIESEC in India (M.A.H.E.)** — Senior Exchange Operations Manager *Aug 2023 – Jul 2025*

- Led a 10+ member team to a Global No. 1 ranking in incoming exchanges and ran end-to-end recruitment of 300+ candidates across multi-round interviews, scheduling, and logistics.

### PATENTS

- Mars Rover Science Module** – on-site collection & analysis of surface and sub-surface samples – application no. 202541099166
- Mechanical & Motorised Wheelchairs for Advanced Motion Capabilities for Impaired Users** — application no. 202441062571.